

Zihan Tang



✉ tangzh23@mails.tsinghua.edu.cn | ☎ +86 17726420782

EDUCATION

Tsinghua University, Beijing, China.

Bachelor of Science in Mathematics and Applied Mathematics, Qiuzhen College Talent Class

- GPA: 3.93/4.00 (Rank: 3/128).

SKILLS

- **Programming:** Python, Lean, C++, L^AT_EX.
- **ML Training:** Large-scale post-training with Verl and AReaL, including PPO/GRPO-style reinforcement learning, SFT, OPD, and model evaluation.
- **Methods:** Proficient in Lean 4 for formal proof development

RESEARCH EXPERIENCE

Pessimistic Verification for Open-Ended Math Questions

07/2025 – 04/2026

ICML 2026

- Co-developed a pessimistic verification method for open-ended mathematical reasoning, improving True Negative Rate from 0.17 to 0.65 on IMO-GradingBench while maintaining high token efficiency.
- Integrated the verifier into a “Solve-Verify-Correct” test-time loop, improving Gemini 3 Pro accuracy from 8.3% to 41.7% on Apex2025.
- arXiv:2511.21522. [Code: <https://github.com/THUNLP-MT/pverify>]

Conditional Entropy Dynamics and Logit-Shift Optimization for RLVR

Ongoing Research Project, AIR, Tsinghua University

05/2026 – Present

- **Learning Dynamics:** Introduced conditional entropy to characterize mode collapse in reasoning models, revealing collapse patterns obscured by global entropy; derived its dynamics under RLVR and separated systematic entropy drift from training variance.
- **Algorithm:** Developed Logit-Shift, a policy-optimization method motivated by the theoretical analysis; trained Qwen3-32B with AReaL on 8 GPUs, improving Hendrycks MATH accuracy from 38% to 84% and consistently outperforming MaxRL.

From Reasoning Chains to Verifiable Subproblems: Curriculum Reinforcement Learning Enables Credit Assignment for LLM Reasoning

09/2025 – 05/2026

Under review at NeurIPS 2026, Co-first Author

- **Algorithm:** Proposed SCRL (Subproblem Curriculum Reinforcement Learning), a curriculum RL frameworks that derives verifiable subproblems from reasoning chains to provide finer-grained credit assignment without external reward models.
- **Theory:** Independently developed a theoretical analysis showing that difficult problems can place outcome-based GRPO in a *gradient dead zone*, where effective learning signals vanish, while subproblem decomposition restores informative gradients with gains increasing with problem difficulty.
- **Results:** Improved over GRPO by +4.1 / +1.9 average points on Qwen3-4B/14B-Base across seven mathematical reasoning benchmarks, including +3.7 pass@1 and +4.6 pass@64 on AIME24/25 and IMO-Bench.
- arXiv:2605.22074

LiveIdeaBench: Benchmarking LLMs on Future Research Idea Forecasting

Under review at EMNLP 2026, Co-first Author 03/2026 – 05/2026
In collaboration with UIUC

- Built a temporally grounded benchmark of 62,572 arXiv cs.ML papers for forecasting future research ideas, and developed a strategy-memory forecasting method achieving Hit@5 of 0.471 and the highest Novelty score (0.238) among baselines using the same backbone.

Mathematical Formalization 01/2025 – 07/2025
 Formalized 1,000+ mathematical problems in Lean 4 for NuminaMath-LEAN, including 80+ IMO-level problems.

- INTERESTS
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- Adaptive learning systems
 - Learning dynamics and optimization
 - Test-time computation

HONORS

Comprehensive Excellence Scholarship (University-level First Class)	10/2025
Academic Excellence Scholarship (University-level Second Class)	10/2024
Qiuzhen Scholarship (Premier College Honor)	10/2025
Chinese Mathematical Olympiad, Silver Medal (National Finalist)	2021, 2022
Chinese Girls' Mathematical Olympiad, Gold Medal (Rank 4)	8/2022